

General form of L.E in two variables

$$ax + by = c, \quad a \neq 0 \wedge b \neq 0$$

three variables

$$ax + by + cz = d, \quad a \neq 0 \wedge b \neq 0 \wedge c \neq 0$$

n variables

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

$$\bullet x - y = 1$$

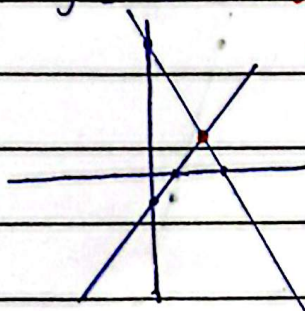
$$2x + y = 6$$

$$x = 1$$

$$y = -1$$

$$x = 3$$

$$y = 6$$



the lines intersect
at one point

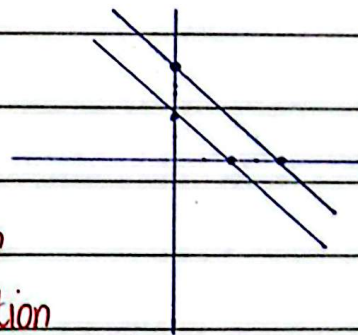
hence, this system
will have one solution
i.e.: unique solution

$$\bullet x + y = 4$$

$$3x + 3y = 6$$

$$x = 4, y = 4$$

$$x = 2, y = 2$$



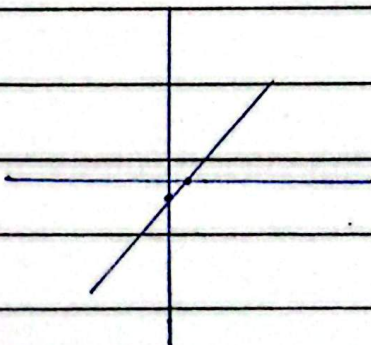
parallel lines have
same slope and
have no solution

$$\bullet 4x - 2y = 1$$

$$16x - 8y = 4$$

$$x = \frac{1}{4}, y = -\frac{1}{2}$$

$$x = \frac{1}{4}, y = -\frac{1}{2}$$



when lines coincide, they have
infinite solution.

System of linear equations in 'n' variables :

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = b_2$$

⋮

$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n = b_m$$

m equations

$$m \neq n, m > n, m < n, m = n$$

its solution is a sequence of numbers

$$= \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix} \rightarrow \text{co-eff matrix}$$

$$= \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} & b_m \end{bmatrix} \rightarrow \text{Augmented matrix}$$

To solve equation, (Basic Elementary operations)

1. Interchange equations
2. Multiply equations by non-zero number
3. Adding or subtracting equations

$$x + y + 2z = 9$$

$$2x + 4y - 3z = 1$$

$$3x + 6y - 5z = 0$$

Its augmented matrix is

convert

$$\begin{bmatrix} 1 & 0 & 0 & a \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & c \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 3 & 6 & -5 & 0 \end{bmatrix}$$

$$R_2 - 2R_1 \rightarrow R_2$$

$$= \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{bmatrix}$$

$$R_3 - 3R_1 \rightarrow R_3$$

$$= \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 3 & -11 & -27 \end{bmatrix}$$

$$\frac{1}{2} R_2 \rightarrow R_2$$

$$= \begin{bmatrix} 1 & 0 & \frac{11}{2} & \frac{35}{2} \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 3 & -11 & -27 \end{bmatrix}$$

$$R_1 - R_2 \rightarrow R_1$$

$$= \begin{bmatrix} 1 & 0 & \frac{11}{2} & \frac{35}{2} \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & -\frac{1}{2} & -\frac{3}{2} \end{bmatrix}$$

$$R_3 - 3R_2 \rightarrow R_3$$

$$= \begin{bmatrix} 1 & 0 & \frac{11}{2} & \frac{35}{2} \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{bmatrix} \quad -2R_3$$

$$= \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix} \quad \begin{array}{l} R_2 + \frac{7}{2}R_3 \rightarrow R_2 \\ R_1 - \frac{11}{2}R_3 \rightarrow R_1 \end{array}$$

$$x = 1, \quad y = 2, \quad z = 3$$

$$SS = (1, 2, 3)$$

1-3 pts.

1-4 pts.

Echelon and Reduced Echelon form :

1. If a row does not contain entirely of zeroes and the first non-zero number in the row is '1', we call this a leading one.
2. If there are any rows that consist entirely of zeros, then they are grouped together at the bottom of the matrix.
3. In any two successive rows that do not consist entirely of zeroes and the leading '1' in the lower row occurs farther to the right than the leading '1' in the higher row.
4. Each column that contains a leading '1' has zero everywhere else in that column.

Echelon' form's example :

$$\begin{bmatrix} 1 & 4 & -3 & 7 \\ 0 & 1 & 6 & 2 \\ 0 & 0 & 1 & 5 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 3 & -1 \\ 0 & 1 & -4 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Reduced Echelon form's examples :

$$\begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

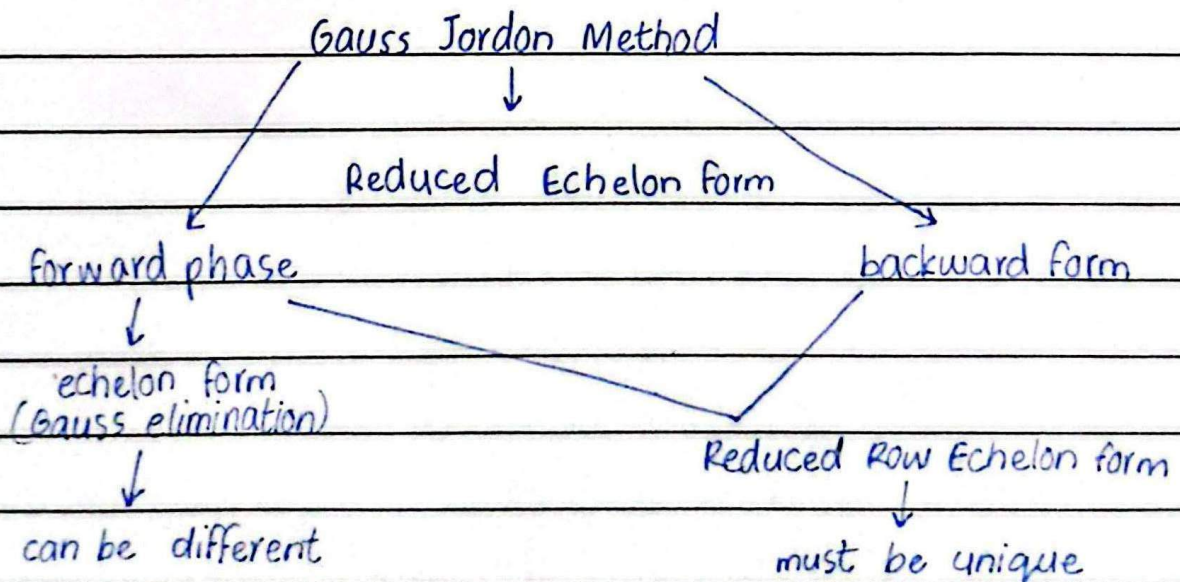
Echelon form example :

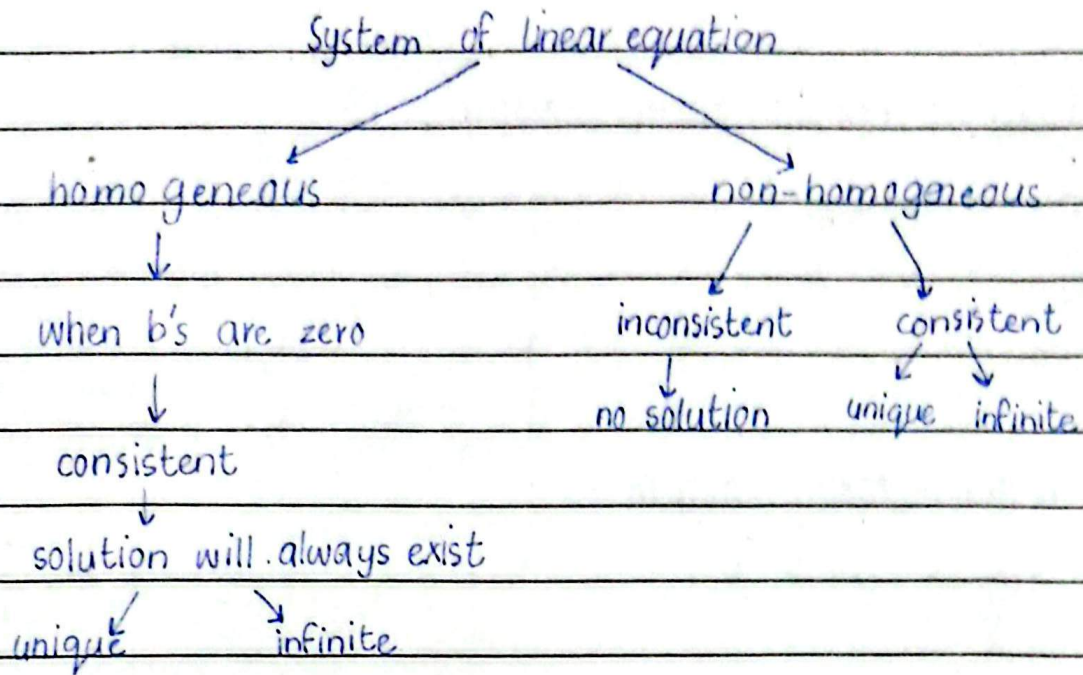
$$\begin{bmatrix} 0 & 0 & 1 & 5 & 6 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

answer will be unique only when no. of variables are equal to no. of equations.

In echelon form, backward substitution is used.

The parameters in solution must be different than the variables involved in the system.





$$\begin{bmatrix} 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -10 & 6 & 12 & 28 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 4 & -10 & 6 & 12 & 28 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix} \quad R_1 \sim R_2$$

$$= \begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix} \quad \frac{1}{2} \times R_1$$

$$= \begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 0 & 0 & 5 & 0 & -17 & -29 \end{bmatrix} \quad R_3 - 2R_1$$

$$= \begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & -\frac{7}{2} & -6 \\ 0 & 0 & 5 & 0 & -17 & -29 \end{bmatrix} \quad -\frac{1}{2} \times R_2$$

$$= \begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & -\frac{1}{2} & -6 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 1 \end{bmatrix} \quad R_3 - 5R_1$$

$$= \begin{bmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & 1 & 0 & -\frac{1}{2} & -6 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{bmatrix} \quad 2R_3$$

Leading variables : x_1, x_3, x_5

Free variables : x_2, x_4

$$x_5 = 2$$

$$x_3 - \frac{1}{2}x_5 = -6$$

$$x_1 + 2x_2 - 5x_3 + 3x_4 - 6x_5 = 14$$

$$x_3 - \frac{1}{2}(2) = -6$$

$$x_3 = 1$$

$$x_1 = -2x_2 + 5x_3 - 3x_4 - 6x_5 + 14$$

$$x_1 = -2x_2 + 5(1) - 3x_4 - 6(2) + 14$$

$$x_1 = -2x_2 - 3x_4 + 7$$

Let $x_2 = s$, $x_4 = t$ where $s, t \in \mathbb{R}$

$$x_1 = -2s - 3t + 7$$

$$x_2 = s$$

$$x_3 = 1$$

$$x_4 = t$$

$$x_5 = 2$$

$$x_1 + x_3 = 0$$

$$x_1 + 4x_2 - 3x_3 = 0$$

$$3x_1 + 6x_2 - 5x_3 = 0$$

$$= \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 4 & -3 & 0 \\ 3 & 6 & -5 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 4 & -4 & 0 \\ 3 & 6 & -5 & 0 \end{bmatrix} \quad \begin{matrix} R_2 - R_1 \\ R_3 - 3R_1 \end{matrix}$$

$$= \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 4 & -4 & 0 \\ 0 & 6 & -8 & 0 \end{bmatrix} \quad R_3 - 3R_1$$

$$= \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 6 & -8 & 0 \end{bmatrix} \quad \begin{matrix} R_2 \times \frac{1}{4} \\ R_3 - 6R_2 \end{matrix}$$

$$= \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & -2 & 0 \end{bmatrix} \quad R_3 - 6R_2$$

$$= \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad \begin{matrix} R_3 \times -\frac{1}{2} \\ R_2 + R_3 \end{matrix}$$

$$= \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad R_2 + R_3$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} R_1 - R_3$$

$$x_1 = 0$$

$$x_2 = 0$$

$$x_3 = 0$$

$$SS = (0, 0, 0)$$

Matrices and Matrix operations :

Arrangements of numbers in form of rows and columns enclosed by squared brackets.

Size of matrices: $[m \times n]$

Let A be any matrix then;

size of $A = \text{Rows} \times \text{Columns}$

Types of matrices :-

Row Matrix / vector :

only one row i.e. $[m=1]$ e.g. $[a_1 a_2 a_3 \dots a_n]_{1 \times n}$

Column Matrix :

only one column. e.g. $\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$

Square Matrix :

If $m=n$, then matrix is square.

e.g. $\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$, $[7]$

Identity Matrix :

Let A be a square matrix then $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$,

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

zero matrix :

All entries are zero. e.g: $[0]$, $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Equal Matrices :

Let A and B be any two matrices then $A=B$ if A and B have same size and same/equal corresponding entries.

e.g: $A = \begin{bmatrix} 1 & 2 \\ x & 5 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$, $A=B$ if $x=3$.

Scalar Matrix :

At least one diagonal entry should be non zero.

Scalar Multiplication:

Let $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 4 \end{bmatrix}$ and $c = 3$ ↑ scalar multiple

$$cA = 3A = 3 \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 6 & 3 \\ 6 & 3 & 12 \end{bmatrix}$$

* $a_{ij} = (A)_{ij} \rightarrow$ entry at i th row and j th column.

Addition and Subtraction:

Matrices must have same size. Let A and B be any two matrices then $A \pm B$ is possible if $|A| = |B|$

Multiplication: columns of 1st one = rows of 2nd one

$$\begin{matrix} \downarrow & & \downarrow \\ A = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} & , & B = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \\ m \times r & & r \times n \end{matrix}$$

Let A and B be two matrices, AB is possible if columns of A and rows of B are equal.
 $m \times n$ is the order of resultant matrix.

$AB \neq BA$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 4 \end{bmatrix}_{2 \times 3}, \quad B = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 1 \end{bmatrix}_{3 \times 3}$$

$$AB = \begin{bmatrix} 1 \times 1 + 2 \times 1 + 3 \times 1 & 1 \times 2 + 2 \times 4 + 3 \times 1 & 1 \times 1 + 2 \times 1 + 3 \times 1 \\ 1 \times 1 + 1 \times 1 + 4 \times 1 & 1 \times 2 + 1 \times 4 + 4 \times 1 & 1 \times 1 + 1 \times 1 + 4 \times 1 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 13 & 6 \\ 6 & 10 & 6 \end{bmatrix}_{2 \times 3}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1r} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2r} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & a_{i3} & \dots & a_{ir} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mr} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1j} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & b_{2j} & \dots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_{r1} & b_{r2} & \dots & b_{rj} & \dots & b_{rn} \end{bmatrix}$$

$$(AB)_{ij} = a_{ij} = [a_{i1} \dots a_{ir}] \begin{bmatrix} b_{1j} \\ \vdots \\ b_{rj} \end{bmatrix} \rightarrow \text{Row column rule for matrix multiplication}$$

Partitioned matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

$$A_{11} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

Row partition or Column partition

Matrix Multiplication [column by column]

jth column vector of $AB = A[\text{jth column of } B]$

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix}$$

$$2^{\text{nd}} \text{ column of } AB = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 7 \end{bmatrix} = \begin{bmatrix} 27 \\ -4 \end{bmatrix}$$

$$4^{\text{th}} \text{ column of } AB = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 3+2+8 \\ 6+6+0 \end{bmatrix} = \begin{bmatrix} 13 \\ 12 \end{bmatrix}$$

Matrix Multiplication [Row by Row]:

i^{th} row of $AB = [i^{\text{th}} \text{ row of } A] B$

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix}$$

$$2^{\text{nd}} \text{ row of } AB = [2 \ 6 \ 0] \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 8+0+0 & 2-6+0 & 8+18+0 & 6+6+0 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & -4 & 26 & 12 \end{bmatrix}$$

Linear combination:

If $A_1, A_2, A_3, \dots, A_r \rightarrow$ matrices of same size

and $c_1, c_2, c_3, \dots, c_r \rightarrow$ scalars

then, $A_1c_1 + A_2c_2 + A_3c_3 + \dots + A_rc_r$ is L.C of $A_1, A_2, A_3, \dots, A_r$

Matrix Product as Linear Combination:

$$\begin{bmatrix} -1 & 3 & 2 \\ 1 & 2 & -3 \\ 2 & 1 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ -9 \\ -3 \end{bmatrix}$$

$$2 \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} - 1 \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 2 \\ -3 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -9 \\ -3 \end{bmatrix}$$

Columns of AB as L.C :

$$AB = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix} = \begin{bmatrix} 12 & 27 & 30 & 13 \\ 8 & -4 & 26 & 12 \end{bmatrix}$$

$$\begin{bmatrix} 12 \\ 8 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$$

Column row expression :

$A \rightarrow m \times r$ matrix $\rightarrow$ partitioned into r columns $c_1, c_2, c_3, \dots$

$B \rightarrow r \times n$ matrix $\rightarrow$ partitioned into r rows $r_1, r_2, \dots, r_r$

$$\text{Then, } AB = c_1 r_1 + c_2 r_2 + \dots + c_r r_r$$

Algebraic properties of matrices :

Properties of Matrix Arithmetic

- $A+B = B+A$ (commutative law of matrix addition)
- $A+(B+C) = (A+B)+C$ (Associative law of matrix addition)
- $A(BC) = (AB)C$ (Associative law for matrix mul.)
- $A(B+C) = AB+AC$ (left distributive law)
- $(B \pm C)A = BA \pm BC$ (Right distributive law)
- $a(B \pm C) = aB \pm aC$
- $(a \pm b)c = ac \pm bc$
- $a(bc) = (ab)c$
- $a(BC) = (aB)C = B(aC)$

Note: order matters in matrix multiplication

$$AB \neq BA \text{ (not always true)}$$

* AB may be defined but not BA . [$A_{2 \times 3}$ and $B_{3 \times 4}$]

* Both defined but have different size. [$A_{2 \times 3}, B_{3 \times 2}$]

* Both defined and also have same size but different entities.

$$A = \begin{bmatrix} -1 & 0 \\ 2 & 3 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$$

$$AB = \begin{bmatrix} -1 & -2 \\ 11 & 4 \end{bmatrix}, BA = \begin{bmatrix} 3 & 6 \\ -3 & 0 \end{bmatrix}$$

$$\Rightarrow AB \neq BA$$

* If $ab = ac$ and $a \neq 0$ then $b = c$

$\Rightarrow$ this property is not true for matrices.

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}, C = \begin{bmatrix} 2 & 5 \\ 3 & 4 \end{bmatrix}$$

$$AB = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}, AC = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$$

* If $ab \neq 0$ then $a = 0$ or $b = 0$

$\Rightarrow$ this property is not true for matrices.

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}, B = \begin{bmatrix} 3 & 1 \\ 0 & 0 \end{bmatrix}$$

$$AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Inverse of a matrix:

Conditions:

$A \rightarrow$ square matrix then if a matrix B s.t

$$AB = BA = I$$

$\Rightarrow A$ is inverse of B

$\Rightarrow B$ is inverse of A .

If A is invertible.

$\Rightarrow A$ is non singular matrix.

Determinant $\neq 0$ (singular)

* If B and C are inverse of A , [uniqueness of inverse]
then $B = C$

* The matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is invertible iff $ad - bc \neq 0$

$$\Rightarrow A^{-1} = \frac{\text{Adj } A}{|A|} = \frac{1}{ad - bc} \cdot \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Q. $A = \begin{bmatrix} 6 & 1 \\ 5 & 2 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 2 \\ 3 & -6 \end{bmatrix}$

$$|A| = 6 \times 2 - 5 \times 1$$

$$= 12 - 5 = 7$$

$$|B| = (-1)(-6) - (3)(2)$$

$$= 6 - 6 = 0$$

$$\text{Adj. of } A = \begin{bmatrix} 2 & -1 \\ -5 & 6 \end{bmatrix}$$

$\therefore |B| = 0$, B^{-1} does not exist.

$$A^{-1} = \frac{1}{7} \begin{bmatrix} 2 & -1 \\ -5 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{2}{7} & -\frac{1}{7} \\ -\frac{5}{7} & \frac{6}{7} \end{bmatrix}$$

^{linear system}
Solution of system by matrix inversion:

$$ax + by = u$$

$$cx + dy = v$$

$$AX = b$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix}, b = \begin{bmatrix} u \\ v \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$$

Let A is invertible

$A^{-1} \rightarrow$ exists

$$A^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1}$$

$$\Rightarrow A^{-1}AX = A^{-1}b$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} \begin{bmatrix} u \\ v \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} \begin{bmatrix} u \\ v \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{du-bv}{ad-bc} \\ \frac{-cu+av}{ad-bc} \end{bmatrix}$$

$$\boxed{x = \frac{du-bv}{ad-bc}}, \quad \boxed{y = \frac{av-cu}{ad-bc}}$$

If A is invertible and B is invertible then

AB is invertible and also $(AB)^{-1} = B^{-1}A^{-1}$

$\Rightarrow A$ and B have same size

Power of a matrix :

* $A^0 = I$ and $A^n = AAA \dots A$ (n factors) (A is square)

* If A^{-1} exists then

$$A^{-n} = (A^{-1})^n = A^{-1}A^{-1} \dots A^{-1} \text{ (n factors)}$$

* $A^r A^s = A^{r+s}$ and $(A^r)^s = A^{rs}$ ($rs = sr$)

* If A^{-1} exists and $n \in \mathbb{Z}^+$ then

1) A is invertible $(A^{-1})^{-1} = A$

2) A^n is invertible and $(A^n)^{-1} = A^{-n} = (A^{-1})^n$

3) kA is invertible for any $k \neq 0 \in \mathbb{R}$ and $(kA)^{-1} = kA^{-1}$

Properties of transpose:

1) $(A^T)^T = A$

2) $(A \pm B)^T = A^T \pm B^T$

3) $(kA)^T = kA^T$

4) $(AB)^T = B^T A^T$

5) $(A^T)^{-1} = (A^{-1})^T$

Properties of zero matrices:

if c is scalar

1) $A + 0 = 0 + A = A$

2) $A - 0 = A$

3) $A - A = A + (-A) = 0$

4) $0A = 0$

5) if $cA = 0$, then $c = 0$ or $A = 0$

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a+b)^2 = a^2 + ab + ba + b^2$$

$$\therefore ab \neq ba$$

$$\text{hence, } (a+b)^2 \neq a^2 + 2ab + b^2$$

Matrix Polynomial:

$$P(x) = x^2 + 3x + 2$$

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix}$$

$$P(A) = A^2 + 3A + 2$$

$$= \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix}^2 + 3 \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix} + 2$$

$$\rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{3} \end{bmatrix} \quad \frac{1}{3} R_2$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad 3R_2 \quad (\text{same as before})$$

$$\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \quad 2R_1 + R_2 \rightarrow R_2$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad -2R_1 + R_2 \rightarrow R_2$$

Equivalent matrices:

Matrices A and B are said to be row equivalent if either (hence each) can be obtained from the other by a sequence of elementary row operations.

Elementary Matrix:

A matrix E is called an elementary matrix if it can be obtained from Identity matrix by performing a single elementary row operation.

e.g:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \downarrow \begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \downarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \downarrow \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \downarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

→ if $E \rightarrow$ elementary matrix from Identity matrix

$A \rightarrow m \times n$

then the product EA is the matrix that results when this same operation is performed on A . e.g:

$$A = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 1 & 4 & 4 & 0 \end{bmatrix}, \quad I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} \quad 3R_1 + R_3 \rightarrow R_3$$

$$EA = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 4 & 4 & 10 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 4 & 4 & 10 & 9 \end{bmatrix} \quad 3R_1 + R_3 \rightarrow R_3$$

$$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad 2R_2 + R_1 \rightarrow R_2$$

$$E_2 E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 3 & 0 & 1 \end{bmatrix}$$

$$E_2 E_1 A = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 5 & -2 & 8 & 15 \\ 4 & 4 & 10 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 5 & -2 & 8 & 15 \\ 4 & 4 & 10 & 9 \end{bmatrix} \quad 2R_2 + R_1 \rightarrow R_2$$

Note : Elementary matrices are always invertible
and their inverse is always an elementary matrix.

homogeneous $\rightarrow$ consistent $\rightarrow$ solution always exist.

Date
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CRYSTAL

$$E = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \quad 3R_2$$

$\downarrow$

$$E^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{3} \end{bmatrix} \quad \frac{1}{3}R_2$$

Equivalent theorem:

Equivalence statement is

if A is $n \times n$ matrix, then the following statements are equivalent, that is all true or all false.

- a) A is invertible.
- b) $AX=0$ has only the trivial solutions.
- c) The reduced Row echelon form of A is I_n .
- d) A is expressible as a product of elementary matrices.

$$a \Rightarrow b \Rightarrow c \Rightarrow d \Rightarrow a$$

Proof:

(a) $\Rightarrow$ (b) Let A is invertible.

$\Rightarrow A^{-1}$ exists.

$$\text{then, } (A^{-1})AX = A^{-1}(0)$$

$$(A^{-1}A)X = A^{-1}(0)$$

$$X = A^{-1}(0)$$

$$X = 0$$

$$X_0 = 0$$

$\Rightarrow AX=0$ has trivial solution $X_0=0$.

(b) $\Rightarrow$ (c) Let $AX=0$ has only trivial solutions

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= 0 \\ \Rightarrow a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= 0 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= 0 \end{aligned} \Rightarrow \left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & 0 \\ a_{21} & a_{22} & \dots & a_{2n} & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & 0 \end{array} \right]$$

As $AX=0$ has only trivial solutions

$$\Rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & \dots & 0 & 0 \\ 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ 0 & 0 & 0 & \dots & 1 & 0 \end{array} \right]$$

$\underbrace{\hspace{10em}}_A \rightarrow$ Augmented

$$A = []$$

$\downarrow$
after k operations
how many
elementary matrices

$\Rightarrow$ R.R.E.F of A is In

(c) $\Rightarrow$ (d) Let R.R.E.F of A is In

A is reduced to In after some finite seq.
of row operations then

$$\Rightarrow E_k \dots E_3 E_2 E_1 A = I_n \quad \text{--- ①}$$

As elementary matrices are invertible,
this implies that

$$\Rightarrow E_k^{-1} \dots E_3^{-1} E_2^{-1} E_1^{-1} \text{ also exist.}$$

Multiply b/s of eq. ① on the left

successively by $E_k^{-1} \dots E_2^{-1} E_1^{-1}$, we obtain

$$E_k^{-1} E_k \dots E_2 E_1 A = E_k^{-1} I_n$$

$$E_{k-1}^{-1} E_k \dots E_2 E_1 A = E_{k-1}^{-1} E_k^{-1} I_n$$

$$A = E_1^{-1} E_2^{-1} \dots E_k^{-1}$$

$$A^{-1} = E_k E_{k-1} \dots E_2 E_1$$

(d) $\Rightarrow$ (a) Given $A = E_1^{-1} E_2^{-1} \dots E_k^{-1}$

as product of all invertible matrices

is also invertible. (Th. 1.4.6)

$\Rightarrow A$ is invertible.

Inversion Algorithm:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$$

$$[A|I] = \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right]$$

$$= \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array} \right] \begin{array}{l} R_2 - 2R_1 \rightarrow R_2 \\ R_3 - R_1 \rightarrow R_3 \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 9 & 5 & -2 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array} \right] \begin{array}{l} R_1 - 2R_2 \rightarrow R_1 \\ R_3 + 2R_2 \rightarrow R_3 \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 9 & 5 & -2 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] -R_3$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \begin{array}{l} R_2 + 3R_3 \rightarrow R_2 \\ R_1 - 9R_3 \rightarrow R_1 \end{array}$$

$$A^{-1} = \begin{pmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{pmatrix}$$

What condition must b_1, b_2 and b_3 satisfy in order for the system of L.E.

$$x_1 + x_2 + 2x_3 = b_1$$

$$x_1 + x_3 = b_2$$

$$2x_1 + x_2 + 3x_3 = b_3$$

$$[A|b] = \left[\begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 1 & 0 & 1 & b_2 \\ 2 & 1 & 3 & b_3 \end{array} \right]$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 0 & -1 & -1 & b_2 - b_1 \\ 0 & -1 & -1 & b_3 - 2b_1 \end{array} \right] \begin{array}{l} R_2 - R_1 \rightarrow R_2 \\ R_3 - 2R_1 \rightarrow R_3 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 0 & -1 & -1 & b_2 - b_1 \\ 0 & 0 & 0 & b_3 - b_2 - b_1 \end{array} \right] R_3 - R_2 \rightarrow R_3$$

For consistency:

$$b_3 - b_2 - b_1 = 0$$

$$\boxed{b_3 = b_1 + b_2}$$

$$\Rightarrow X = A^{-1}b$$

Diagonal matrix: At least one diagonal entity must be non zero and all other entities must be zero.

e.g: $\begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix}$

inverse exists only when all diagonal entities are non zero. Inverse is reciprocal of diagonal entities.

Triangular matrix : must be square

Upper (Echelon form)

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

lower

$$A = \begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

transpose of upper = lower and vice versa

product of 2 upper = upper

product of 2 lower = lower

inverse of upper = upper

inverse of lower = lower

inverse exists iff diagonal entities are all non zero

Symmetric matrix :

If A is any square matrix if $(A^T) = A$.

specific entity $\leftarrow (A)_{ij} = (A)_{ji}$ $[A]_{ij} \rightarrow$ whole matrix

$$\rightarrow (A^t)^t = A$$

$$\star (AB)^t = (AB) \text{ iff } AB = BA$$

$\star$ If inverse of A exists then

$$AA^t \text{ and } A^tA \rightarrow \text{invertible}$$

Determinants :

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

★ Minor of a_{ij} represented by M_{ij}

$$M_{11} = \begin{vmatrix} e & f \\ h & i \end{vmatrix} = ei - fh$$

★ Cofactor of a_{ij} represented by C_{ij}

$$C_{ij} = (-1)^{i+j} M_{ij}$$

$$C_{11} = M_{11}$$

$$C_{12} = -M_{12}$$

$$A = \begin{bmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{bmatrix}$$

$$M_{11} = \begin{vmatrix} 5 & 6 \\ 4 & 8 \end{vmatrix} = 40 - 24 = 16 \Rightarrow C_{11} = 16$$

$$M_{12} = \begin{vmatrix} 2 & 6 \\ 1 & 8 \end{vmatrix} = 16 - 6 = 10 \Rightarrow C_{12} = -10$$

$$M_{13} = \begin{vmatrix} 2 & 5 \\ 1 & 4 \end{vmatrix} = 8 - 5 = 3 \Rightarrow C_{13} = 3$$

$$M_{21} = 8 + 16 = 24 \Rightarrow C_{21} = -24$$

$$M_{22} = 24 + 4 = 28 \Rightarrow C_{22} = 28$$

$$M_{23} = 11 \Rightarrow C_{23} = -11$$

$$M_{31} = 26 \Rightarrow C_{31} = 26$$

$$M_{32} = 26 \Rightarrow C_{32} = -26$$

$$M_{33} = 13 \Rightarrow C_{33} = 13$$

$$C_{ij} = \begin{bmatrix} 16 & -10 & 3 \\ -24 & 28 & -11 \\ 26 & -26 & 13 \end{bmatrix} \quad A = \begin{bmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 3 \end{bmatrix}$$

$$\det(A) = 3(16) + 1(-10) - 4(3) \\ = 26$$

Take Any rows from A and its corresponding row from cofactors matrix then multiply and add its all entries, it will give determinant. But if rows are diff. i.e. 1st row from A and 2nd row from C_{ij}, answer will always be zero. (same for columns)

$$\star \quad A^{-1} = \frac{\text{Adj } A}{|A|} = \frac{(C_{ij})^t}{|A|}$$

$$A^{-1} = \frac{1}{26} \begin{bmatrix} 16 & -24 & 26 \\ -10 & 28 & -26 \\ 3 & -11 & 13 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{16}{26} & \frac{-24}{26} & 1 \\ \frac{-10}{26} & \frac{28}{26} & -1 \\ \frac{3}{26} & \frac{-11}{26} & \frac{13}{26} \end{bmatrix}$$

the determinant of triangular and diagonal matrix comes by multiplying all entries that are diagonal.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 9 \end{bmatrix}$$

$$\det(A) = (1)(2)(4) = 8$$

$$\det(A) = (1)(1)(9) = 9$$

★ if one entire row or entire column is zero, determinant is zero.

★ A matrix which has proportional rows or columns, its determinant is zero.

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \\ 1 & 0 & 1 \end{bmatrix} \quad R_1 = 2R_2$$

$$\star \quad A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$B = \begin{bmatrix} ka & kb & kc \\ d & e & f \\ g & h & i \end{bmatrix} \quad kR_1$$

$$\det(B) = k \det(A)$$

$$B = \begin{bmatrix} d & e & f \\ a & b & c \\ g & h & i \end{bmatrix} \quad R_1 \sim R_2$$

$$\det(B) = -\det(A)$$

$$B = \begin{bmatrix} a+kd & b+ke & c+kf \\ d & e & f \\ g & h & i \end{bmatrix} \quad R_1 + kR_2 \rightarrow R_1$$

$$\det(B) = \det(A)$$

$$A = \begin{bmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{bmatrix}$$

$$\det(A) = \begin{vmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{vmatrix}$$

$$= - \begin{vmatrix} 1 & 4 & 8 \\ 2 & 5 & 6 \\ 3 & 1 & -4 \end{vmatrix} \quad R_1 \sim R_3$$

$$= - \begin{vmatrix} 1 & 4 & 8 \\ 0 & -3 & -10 \\ 0 & -11 & -28 \end{vmatrix} \quad \begin{array}{l} R_2 - 2R_1 \\ R_3 - 3R_1 \end{array}$$

$$= -(-3) \begin{vmatrix} 1 & 4 & 8 \\ 0 & 1 & \frac{10}{3} \\ 0 & -11 & -28 \end{vmatrix} \quad \begin{array}{l} -1 R_2 \\ 3 \end{array}$$

$$= 3 \begin{vmatrix} 1 & 4 & 8 \\ 0 & 1 & \frac{10}{3} \\ 0 & 0 & \frac{26}{3} \end{vmatrix} \quad R_3 + 11R_2$$

$$= 3 (1)(1) \left(\frac{26}{3} \right)$$

$$= 26$$

$$I_n \xrightarrow{R_1 \sim R_2} E \Rightarrow \det(E) = -1$$

$$I_n \xrightarrow{kR_1} E \Rightarrow \det(E) = k$$

$$I_n \xrightarrow{R_1 + kR_2} E \Rightarrow \det(E) = 1$$

$$A^{-1} = \frac{\text{adj}(A)}{\det(A)}$$

$$C = \begin{bmatrix} C_{11} & \dots & C_{1n} \\ \vdots & & \vdots \\ C_{n1} & \dots & C_{nn} \end{bmatrix}$$

Proof :-

$$A \text{adj}(A) = \det(A)I \quad \because A \text{ is invertible}$$

Consider

$$\begin{aligned} A \text{adj}(A) &= \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} C_{11} & \dots & C_{1j} & \dots & C_{1n} \\ \vdots & & \vdots & & \vdots \\ C_{n1} & \dots & C_{jn} & \dots & C_{nn} \end{bmatrix} = C^T \\ &= \begin{bmatrix} \det(A) & 0 & \dots & 0 \\ 0 & \det(A) & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & \det(A) \end{bmatrix} \end{aligned}$$

$$A \text{adj}(A) = \det(A)I$$

Since A is invertible,

$$A \text{adj}(A) \begin{bmatrix} 1 \\ \vdots \\ \det(A) \end{bmatrix} = I$$

$$A \begin{bmatrix} \text{adj}(A) \\ \det(A) \end{bmatrix} = I$$

$$A^{-1}A \begin{bmatrix} \text{adj}(A) \\ \det(A) \end{bmatrix} = A^{-1}I$$

$$A^{-1} = \frac{\text{adj}(A)}{\det(A)}$$

Important points:

- * $\det(kA) = k^n \det(A)$
- * $\det(A+B) \neq \det(A) + \det(B)$
- * $\det(A) + \det(B) = \det(C) \rightarrow$ if any rows are same in A, B, C
e.g: $A = \begin{bmatrix} 1 & 7 & 5 \\ 2 & 0 & 3 \\ 1 & -1 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 & 7 & 5 \\ 2 & 0 & 3 \\ 7 & 2 & 1 \end{bmatrix},$
- * $\det(AB) = \det(A)\det(B)$
- * $\det(EA) = \det(E)\det(A)$ $C = \begin{bmatrix} 1 & 7 & 5 \\ 2 & 0 & 3 \\ 2 & 4 & 8 \end{bmatrix}$
- * $\det(A^{-1}) = \frac{1}{\det(A)}$

Cramer's rule: (determinant must be non zero)

- * If Determinant is not zero, unique solution.
 - * If Determinant is zero, infinite or no solution.
- $\rightarrow$ The order of matrix must be same.

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, b = \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

$$x_1 = \frac{\det(A_1)}{\det(A)}, x_2 = \frac{\det(A_2)}{\det(A)}, x_3 = \frac{\det(A_3)}{\det(A)}$$

$$A_1 = \begin{bmatrix} p & b & c \\ q & e & f \\ r & h & i \end{bmatrix}, A_2 = \begin{bmatrix} a & p & c \\ d & q & f \\ g & r & i \end{bmatrix}, A_3 = \begin{bmatrix} a & b & p \\ d & e & q \\ g & h & r \end{bmatrix}$$

Equivalence theorem:

If A is nxn matrix, then the following statements are equivalent:

- a) A is invertible.
- b) $AX=0$ has only trivial solution.
- c) The R.R.E.F of A is I_n .
- d) A can be expressed as a product of elementary matrices.
- e) $AX=b$ is consistent for every $n \times 1$ matrix b .
- f) $AX=b$ has exactly one solution for every $n \times 1$ matrix b .
- g) $\det(A) \neq 0$

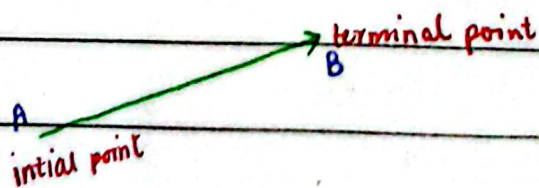
Inverse of equivalence theorem:

- a) A is non invertible (singular).
- b) $AX=0$, non trivial.
- c) R.R.E.F of $A \neq I_n$ (one row is 0)
- d) A cannot be expressed as product of elementary matrices.
- e) $AX=b$ is inconsistent for at least one $n \times 1$ b
- f) $AX=b$ does not have unique solution.
- g) $\det(A) = 0$

CH:03

Euclidean vector space:

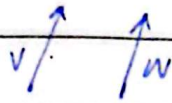
Vectors in 2-space, 3-space and n -space.



$$V = \vec{AB}$$

Equal/Equivalent vectors:

parallel vectors are equal. same length and direction.



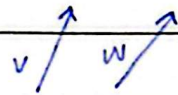
magnitude must be equal as well.

zero vector:

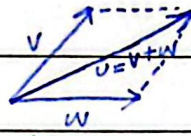
magnitude = 0 or initial point = terminal point

Rules for vector addition:-

1. Parallelogram rule



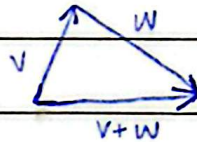
combine tails



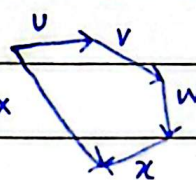
draw a diagonal

diagonal is sum.

2. Triangle rule

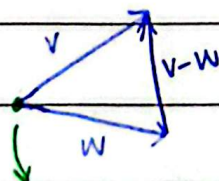
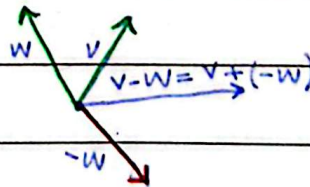
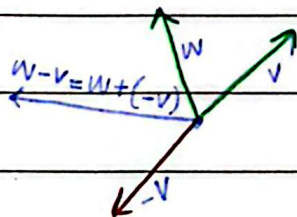


$u + v + w + x$



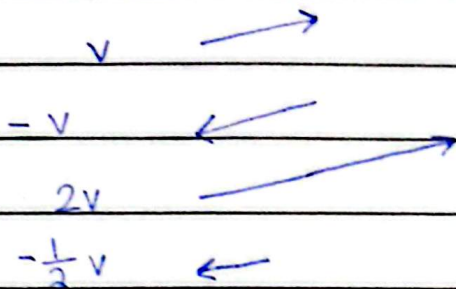
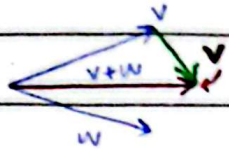
for more variables

Subtraction:

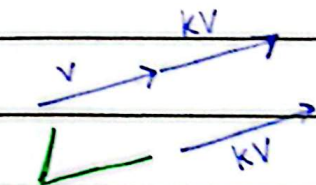


combine tails

Translation of vectors:



Collinear vectors: multiples of each other



v and kv are collinear as they are parallel and multiples of each other

Parallelepiped

$$v = (v_1, v_2, \dots, v_n)$$

$$w = (w_1, w_2, \dots, w_n)$$

$$v \pm w = (v_1 \pm w_1, v_2 \pm w_2, \dots, v_n \pm w_n)$$

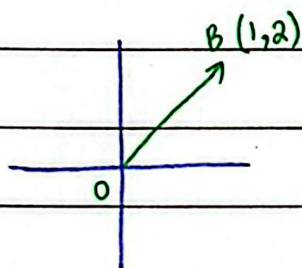
$$kv = (kv_1, kv_2, \dots, kv_n)$$

$$-1v = (-v_1, -v_2, \dots, -v_n)$$

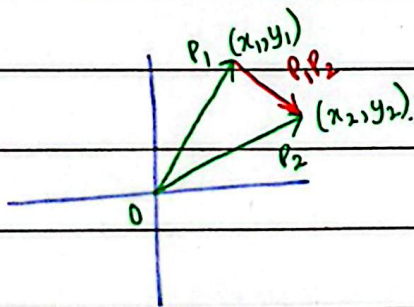
$$0 \cdot \vec{v} = \vec{0}$$

If u, v and w are three vectors ^{in $\mathbb{R}^n$} and k and m are scalars:

- 1) $u+v = v+u$
- 2) $u+(v+w) = (u+v)+w$
- 3) $u+0 = 0+u = u$
- 4) $u+(-u) = 0$
- 5) $k(u+v) = ku + kv$
- 6) $(k+m)u = ku + mu$
- 7) $k(mu) = km(u)$
- 8) $1u = u$



$$\vec{OB} = (1, 2)$$



$$\begin{aligned}\vec{P_1P_2} &= \vec{OP_2} - \vec{OP_1} = (x_2, y_2) - (x_1, y_1) \\ &= (x_2 - x_1, y_2 - y_1)\end{aligned}$$

Linear combinations:

$$v_1 = (1, 2, 3)$$

$$v_2 = (1, 4, 7)$$

$$v_3 = (2, 1, 4)$$

$$2v_1 + 3v_2 + v_3 = u \rightarrow \text{L.C of } u$$

$$u = (7, 17, 31)$$

let w be a vector in $\mathbb{R}^n$ and $v_1, v_2, v_3, \dots, v_n$ are in $\mathbb{R}^n$ with scalars $a_1, a_2, a_3, \dots, a_n$, then w is said to be L.C of $v_1, v_2, v_3, \dots, v_n$.

$$w = a_1 v_1 + a_2 v_2 + a_3 v_3 + \dots + a_n v_n$$

$$u = k_1 v_1 + k_2 v_2 + k_3 v_3$$

$$(7, 17, 31) = k_1(1, 2, 3) + k_2(1, 4, 7) + k_3(2, 1, 4)$$

$$(7, 17, 31) = (k_1 + k_2 + 2k_3, 2k_1 + 4k_2 + k_3, 3k_1 + 7k_2 + 4k_3)$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 7 \\ 2 & 4 & 1 & 17 \\ 3 & 7 & 4 & 31 \end{array} \right]$$

↓

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1 \end{array} \right]$$

$$v = (a, b, c) \rightarrow \text{comma-delimited form}$$